

CS 357 – Numerical Methods 1

We will **not** use the Zoom chat for questions

Instead, all questions should be posted on **Discord**

Course staff will be answering
questions in real-time during class

Join Discord



Meeting Mariana



Teaching Associate Professor

<https://mfsilva.web.illinois.edu>

mfsilva@illinois.edu

Research Area: [Computers and Education](#)

- Started teaching at UIUC Sp 2012
- Taught 10 different courses
- Teaching CS 357 since Sp 2018

My professional interests

- Computer-based assessments and testing
- Fostering collaborations in the classroom
- Leveraging Gen-AI for grading, feedback and question generation
- Development and adoption of technological innovations for large-scale teaching
 - Co-founded [PrairieLearn Inc](#) in 2021



Meeting Luke



Professor

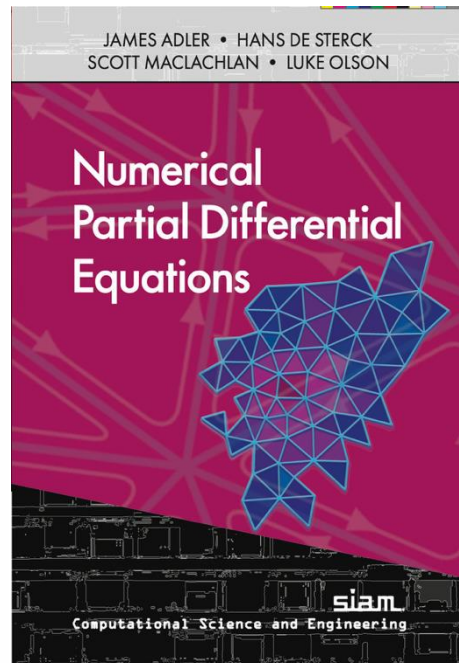
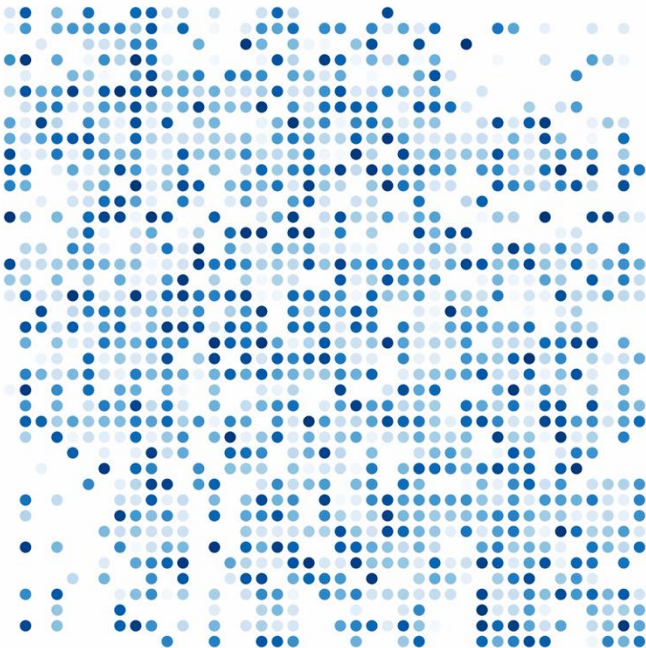
[https://lukeo.cs.illinois.edu/
lukeo@illinois.edu](https://lukeo.cs.illinois.edu/lukeo@illinois.edu)

Research Area: Scientific Computing

- Started at UIUC in 2005
- Courses:
 - CS450: Numerical Analysis
 - CS555: Numerical PDEs
 - CS556: Iterative Methods for Sparse Matrices
 - CS598sml: Sci. Machine Learning

My professional interests

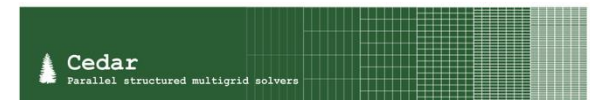
- Iterative and Multigrid Methods for solving large linear systems
- Performance modeling
- Finite Element methods for PDES



PyAMG
Algebraic Multigrid Solvers in Python



RAPtor: parallel algebraic multigrid



CUSP

Meet Course Staff

Get support

- All communication will happen via Discord. NO EMAILS!
- Check Discord often (specially the #announcement channel)
- Office Hours: in-person and virtual options (**starts on week 2**)

Other important announcements

- Our classroom values respect, diversity, and inclusivity. **Everyone is welcome here!**
- Eating is NOT allowed in classroom. You must eat your lunch before/after class (CIF policy!)
- There will be Course Surveys and Consent Form for extra credit. Announcements via Discord. **First survey on week 3.**

Other opportunities for extra credit

- Start of semester survey: 0.25%
- Research study survey on week 6: 0.25%
- End of semester survey: 0.25%
- Extra credit questions included in group activities: up to 0.27%

Course Website - Syllabus

<https://cs357.cs.illinois.edu/pages/syllabus.html>

Course Assessments

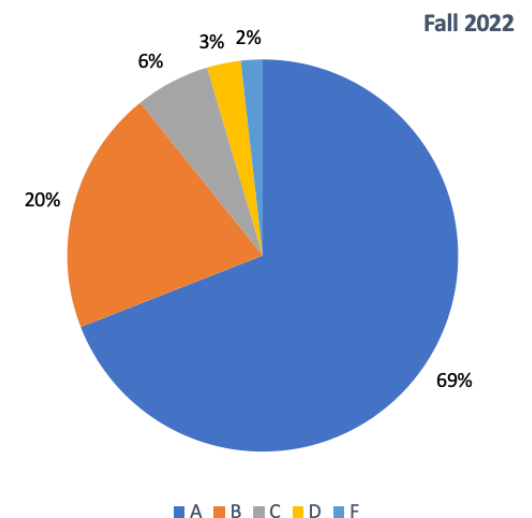
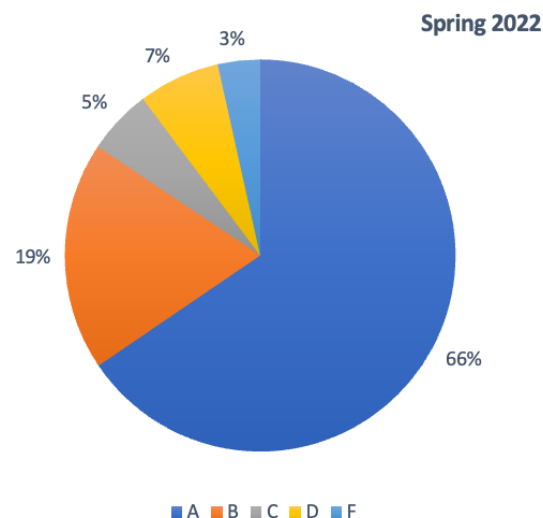
Assessment	Contribution	Info
Final exam	15%	Delivered via PrairieLearn at the CBTF (asynchronously with self-registration).
Bi-weekly quizzes	35%	Delivered via PrairieLearn at the CBTF (asynchronously with self-registration). Read more here .
Homework	25%	Submitted twice weekly via PrairieLearn.
MP	10%	Submitted 5 times during the semester via PrairieLearn.
Group Activity Completion	9%	Weekly group work to be completed on Tuesdays (read more here).
Group Activity Participation	2%	Weekly group work to be completed on Tuesdays (read more here). Participation is computed based on attendance for students in the in-person section and completion of group feedback survey for students in the online section.
Pre-Lecture Activity	4%	Pre-recorded videos available via PrairieLearn on Tuesdays and Thursdays with corresponding short questions

Final Letter Grade Calculation

Grade Point Range

A	[93, 100)
A-	[90, 93)
B+	[87, 90)
B	[83, 87)
B-	[80, 83)
C+	[77, 80)
C	[73, 77)
C-	[70, 73)
D+	[67, 70)
D	[63, 67)
D-	[60, 63)
F	< 60

- We never curve
- We do not round grades
- Up to 1% in extra credit points
- A+ given to top 5% students



Assessments (all in PrairieLearn)

Module 7. Randomness and Monte Carlo Methods

L7

[Monte Carlo Methods](#)

HW7

[Monte Carlo Methods](#)

GA 4

[Using random walk to model stock market](#) 

MP 1

[Machine Problem 1: Numerical Experiments using Monte Carlo](#)

- (Pre-)Lecture
- Homework
- Group Activity
- Machine Problem
- Practice Quiz

Let us know if you are not familiar with PrairieLearn and/or PrairieTest. We can give you a demo during office hours, or schedule a separate appointment.



CS 357 Spring 2026

Numerical Methods I

Syllabus

PrairieLearn

 Discord

Pre-Lectures

L7: Monte Carlo Methods

Total points: 0/3

0%

Assessment is **open** and you can answer questions.
Available credit: 100% (Staff override) ⓘ

Resources for Monte Carlo Methods: [Notes and complete slides](#).

You can also follow along the video lectures using the notebooks from this [workspace](#) .

Question	Value	History	Awarded points
Empty JupyterLab for your personal working area			
L7.1. Your personal JupyterLab	0		—
Random numbers			
L7.2. Video 1: Random numbers	0		—
L7.3. Example 1: Random Number Examples	0		—
L7.4. Clock Generator	1		— /1
L7.5. Properties of Random Number Generators	1		— /1

- Videos + short check-point questions
- Option to read [online textbook](#) and/or slides
- 4 lowest scores will be dropped (4% of final grade)

Homework

HW7: Monte Carlo Methods			
Total points: 0/24		Assessment is open and you can answer questions. Available credit: 100% (Staff override) ⓘ	
		0%	
Question	Value	History	Awarded points
Empty JupyterLab for your personal working area			
HW7.1. Your personal JupyterLab	0		—
Linear Congruential Generators			
HW7.2. Linear Congruential Generators	1		— /3
Monte Carlo Conceptual			
HW7.3. Where to use Monte Carlo	1		— /1
HW7.4. Monte Carlo Convergence Definition	1		— /1

- 6-15 questions (MC, checkbox, fill-in-the-blank, coding)
- 2 lowest scores will be dropped (25% of final grade)

Group Activity (Tuesdays)

- Completed on Tuesdays, open from 12:30pm - 10pm
- Required class attendance for students in Section N (but can still complete the GA until 10pm)
- Section M students can complete the GA anytime on Tuesday, from any location. However support is only provided during class time via Discord.
- Mini-introduction (5-10 min): section M students can join on Zoom
- Lowest two scores will be dropped (9% of final grade)
- Participation points count towards 2% of the final grade
- More about GAs next class

Study Hall (Thursdays)

- Optional attendance
- More examples and review
- Canceled 6 times during the semester (to compensate for quiz scheduling)

Machine Problems

MP 1: Machine Problem 1: Numerical Experiments using Monte Carlo			
Total points: 0/24		Assessment is open and you can answer questions. Available credit: 100% (Staff override) ⓘ	
		0%	
Question	Value	History	Awarded points
Empty JupyterLab for your personal working area			
MP 1.1. Your personal JupyterLab	0		—
Texas Holdem			
MP 1.2. Texas Hold 'em game - introduction	4		— /4
MP 1.3. Dealing hands at random 🔒	4		— /4
MP 1.4. Determining who wins 🔒	4		— /4
MP 1.5. Determining win probabilities 🔒	4		— /4
MP 1.6. Running different numerical experiments 🔒	4		— /4
MP 1.7. Final conclusions 🔒	4		— /4

- Longer coding problems
- Scaffolded
- Appear 5 times in the semester, no drops (10% of final grade)

Quizzes

- 6 quizzes; lowest score is dropped (35% of final grade)
- 8-12 non-coding questions + 2-3 coding questions
- 50-min quizzes all at CBTF (check schedule on course website)
- DRES students should provide letter to CBTF
- We will use PrairieTest to make quiz reservations.
- MAKE YOUR RESERVATION AS SOON AS SLOTS OPEN! We will not provide special overrides because you were not able to find a slot that works for you.
- Practice Quizzes – generated mostly from the same set of questions that will generate the actual quizzes

Exams available for reservations			
Action	Exam	First date	Last date
Make a reservation	CS 357 (Sp23): Quiz 1	2023-01-30 00:01:00 (CST)	2023-02-01 23:59:00 (CST)

Using AI tools in this course

Use of LLM Tools (e.g., ChatGPT, Claude, Gemini)

You are welcome to use Large Language Model (LLM) tools as part of your learning process in the same way you are encouraged to discuss ideas with your peers. LLMs can be helpful for clarifying concepts, giving hints, or offering alternative ways to think about a problem. However, it is essential that you use these tools responsibly. Asking an LLM to solve the assignment for you, or relying on its output without understanding it, does not support your learning. You may end up with correct answers on homework, MPs, or group activities, which together make up 50% of your grade, but you will very likely struggle on quizzes and the final exam, which make up the remaining 50% and must be completed individually without help (you don't get access to these tools inside the CBTF).

Formative assignments (homework, MPs, or group activities) exist to help you learn. You are expected to make sure you understand the material and can solve the problems on your own. LLM tools, peers, and course staff should support your thinking, not replace it. Ultimately, all work you submit must reflect your own understanding, reasoning, and effort. This will help you succeed in the course!

Week 1

Tue, Jan 20, lecture 1

Syllabus and Intro to CS 357

Complete your asynchronous lecture today!

L1: Introduction to CS 357 (NOR FOR CREDIT)

L2: Introduction to Python

Also opening today:

HW1: Linear Algebra Review (NOT FOR CREDIT)

HW2: Introduction to Python

Due today:

Thu, Jan 22, lecture 2

Python Intro + mock group work

Complete your asynchronous lecture today!

L3: Errors, Big-O notation, plots

Also opening today:

HW3: Errors and Big-O

Due today:

In general lectures and HWs will open at 8am Tuesdays and Thursdays.

Week 2

Tue, Jan 27, lecture 3

GA1: Working with Python

Complete your asynchronous lecture today!

L4a: Floating point

Also opening today:

HW4a: Floating point

Q1P: Linear Algebra + Python + Errors

Due today:

HW2: Introduction to Python

Module 2. Python

L2 [Introduction to Python](#)

HW2 [Introduction to Python](#)

GA 1 [Working with Python](#) 

This will be the
GA next Tuesday!

Module 1. Introduction

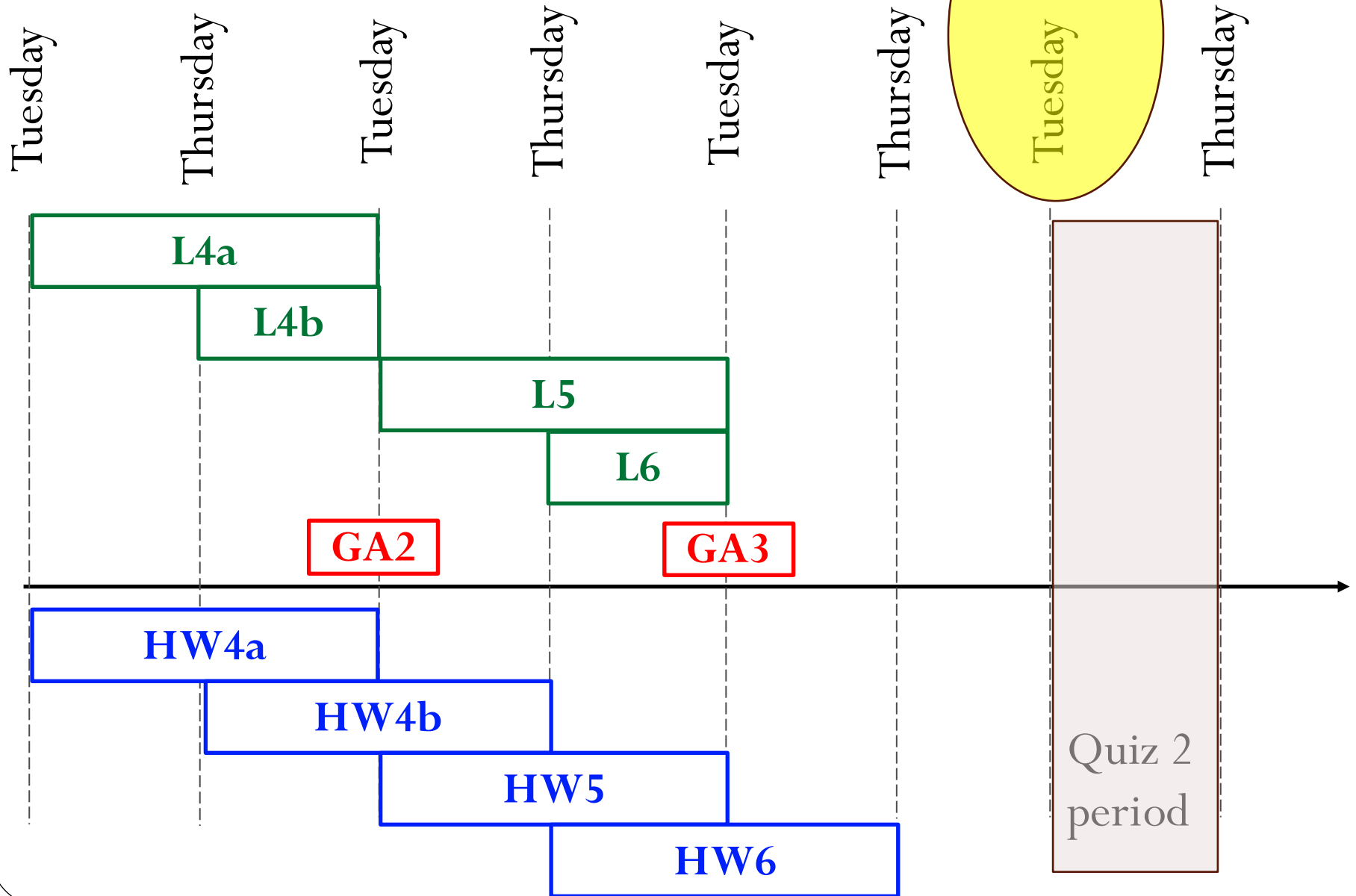
L1 [Introduction to CS 357 \(NOT FOR CREDIT\)](#)

HW1 [Linear Algebra Review \(WILL NOT COUNT FOR COURSE CREDIT\)](#)

GA 0 [Get started with GAs \(NOT FOR CREDIT\)](#) 

GA00 [Workspaces for collaborative learning \(NOT FOR CREDIT\)](#) 

Deadlines



Learning Flow for each Module

1. Complete lectures on Tuesdays and Thursdays as they open
 - Get up to 100% credit by completing them before Tuesday 12pm of the week after they open
 - Get up to 80% credit by completing them before Tuesday midnight of the corresponding quiz week
 - Get up to 50% credit by completing them before last day of classes
2. Complete the Homework
 - Get up to 100% credit by completing them within one week after they open
 - Get up to 96% credit by completing them before Tuesday midnight of the corresponding quiz week
 - Get up to 50% credit by completing them before last day of classes
3. Complete MPs (when available) – similar deadline scheme as HW
4. For additional examples, look at the Demos (not for credit)

Practice Quizzes

1. You are encouraged to start the practice quizzes only after you complete the learning flow above
2. Complete at least one entire practice quiz as if you were taking the quiz (no access to resources, timed, etc). This will give you a good idea of how ready you are.

Assessments		
Module 3. Errors and Big-O		
L3	Errors, Big-O notation, plots	
HW3	Errors and Big-O	
Module 2. Python		
L2	Introduction to Python	
HW2	Introduction to Python	
D2	Demo: Additional Python Tutorial	
Module 1. Introduction		
L1	Introduction to CS 357 (NOT FOR CREDIT)	
HW1	Linear Algebra Review (NOT FOR CREDIT)	
D1	Demo: Intro to Numerical Methods	
GA00	Workspaces for collaborative learning (NOT FOR CREDIT)	

Credit	Start	End
100%	2023-08-24 08:00:01-05 (CDT)	2023-08-29 12:00:00-05 (CDT)
80%	2023-08-24 08:00:01-05 (CDT)	2023-09-05 23:59:59-05 (CDT)
50%	2023-08-24 08:00:01-05 (CDT)	2023-12-06 23:59:59-06 (CST)
0%	2023-08-24 08:00:01-05 (CDT)	—
		None

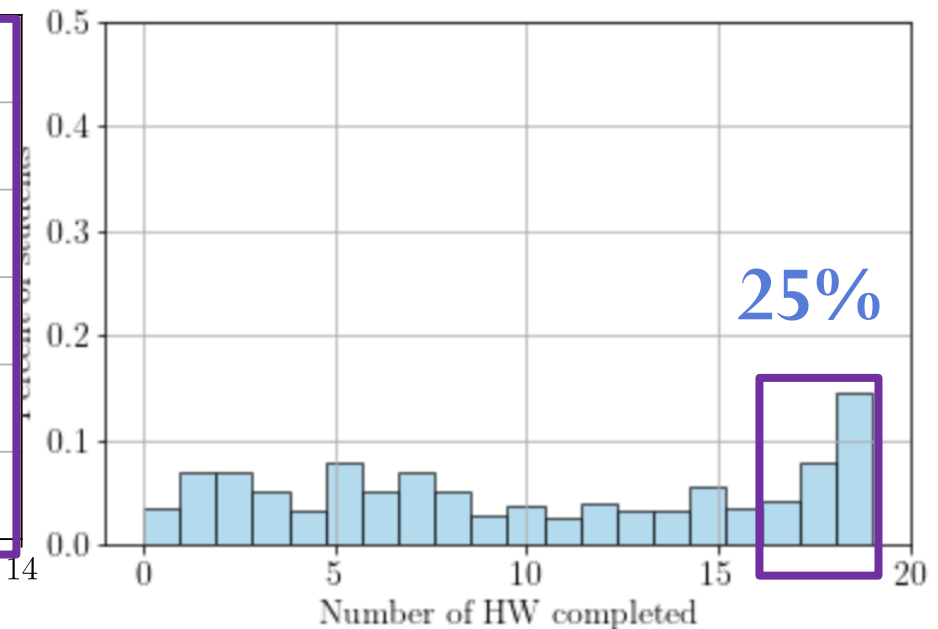
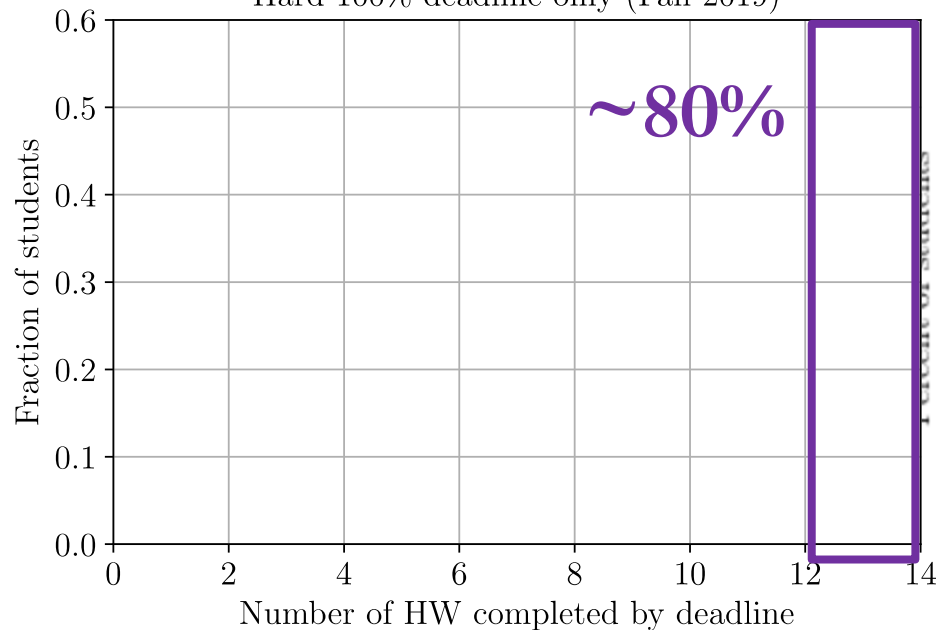
Score	
?	0%
1 ?	Not started
9 ?	0%
9 ?	0%
	0%
	0%
	New instance
	0%
	Not started

What did we learn about flexible deadlines in this class?

With “hard” one-week deadlines

With flexible deadlines

Hard 100% deadline only (Fall 2019)

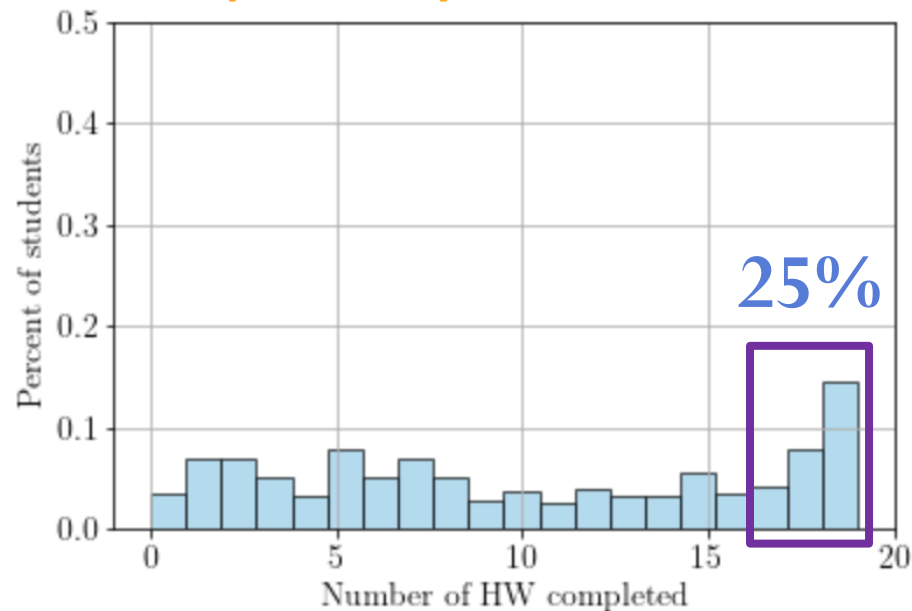


Percent of students who complete most of the HW decreased from ~80% to ~25%!

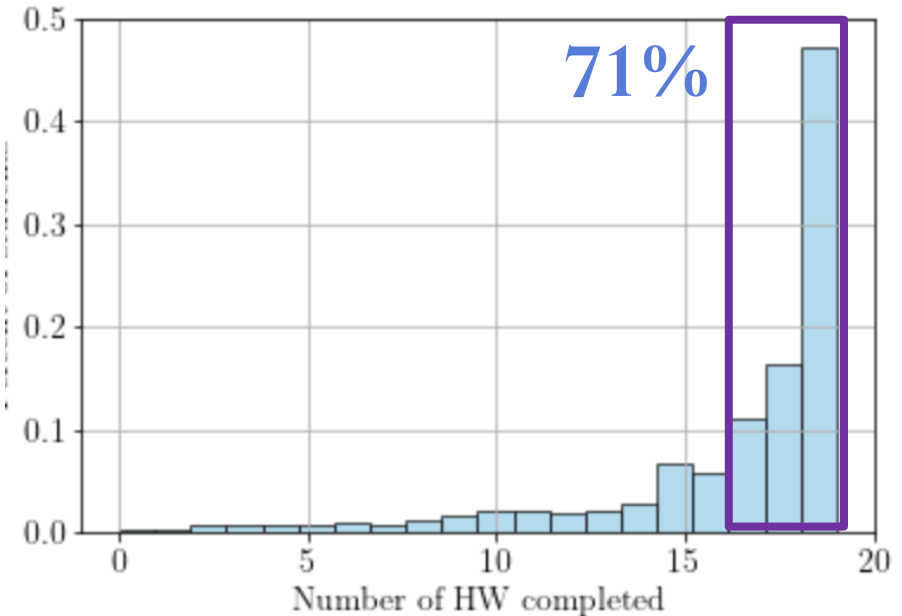
What did we learn about flexible deadlines in this class?

With flexible deadlines

Completion by 100% deadline



Completion by 96% deadline

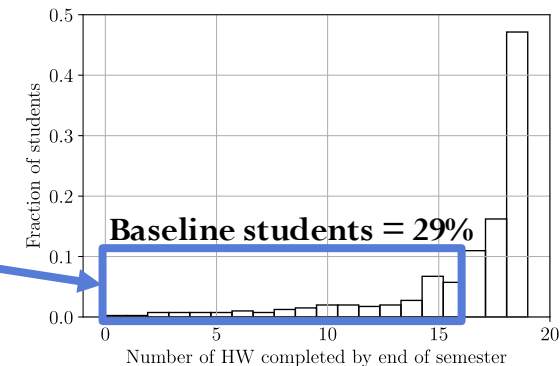


Percent of students who complete most of the HW
Increased to 71% by the quiz date

How HW completion impact quiz performance

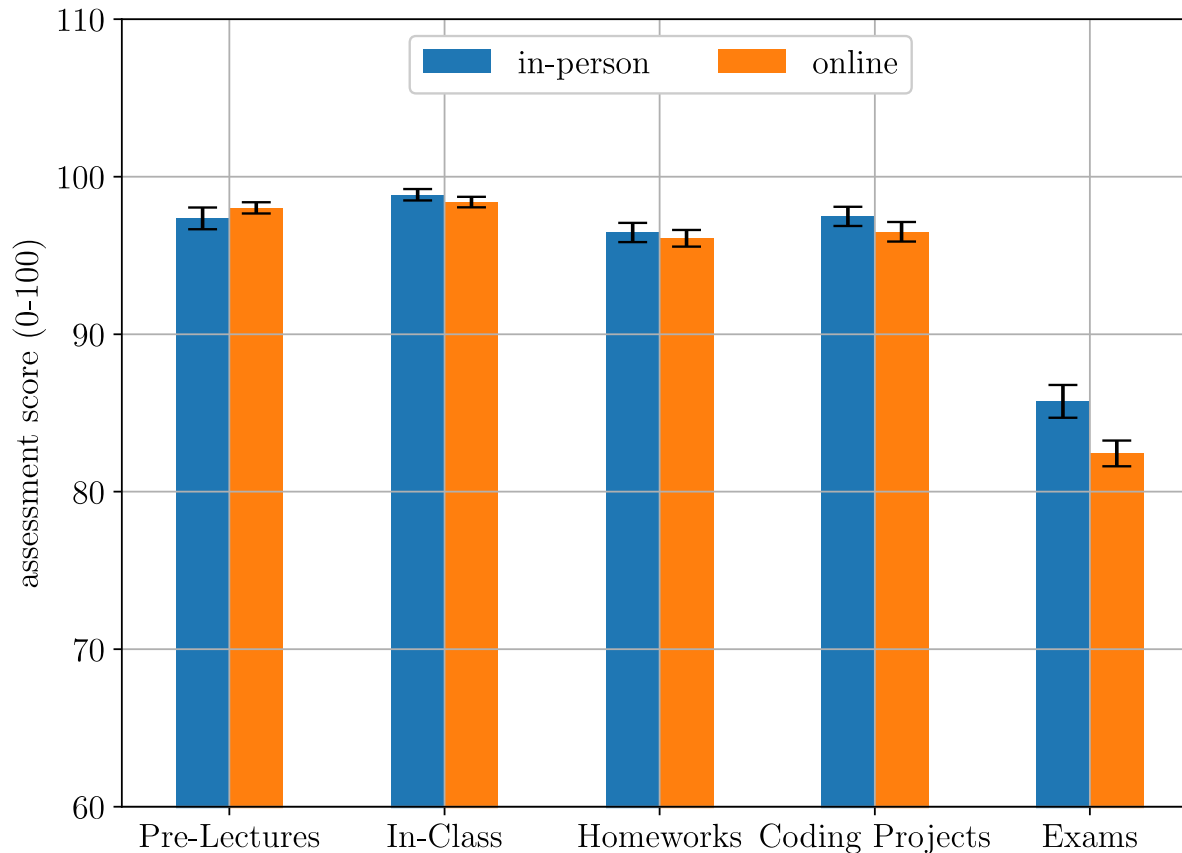
Regression model to fit exam scores including control for GPA

Baseline: students who did not complete HWs by the exam date



Students who complete HWs ...	average score advantage on exams	% students
Do not complete	0% (baseline)	29%
by exam date	14.4%	46%
within one week	18.6%	25%

What do we know about online vs in-person sections in CS 357?



In-person and Online students have similar performance on all “learning” assignments

In-person students have **ON AVERAGE** a small (2.4%) score advantage on quizzes when compared to **online** students

Sense of Belonging

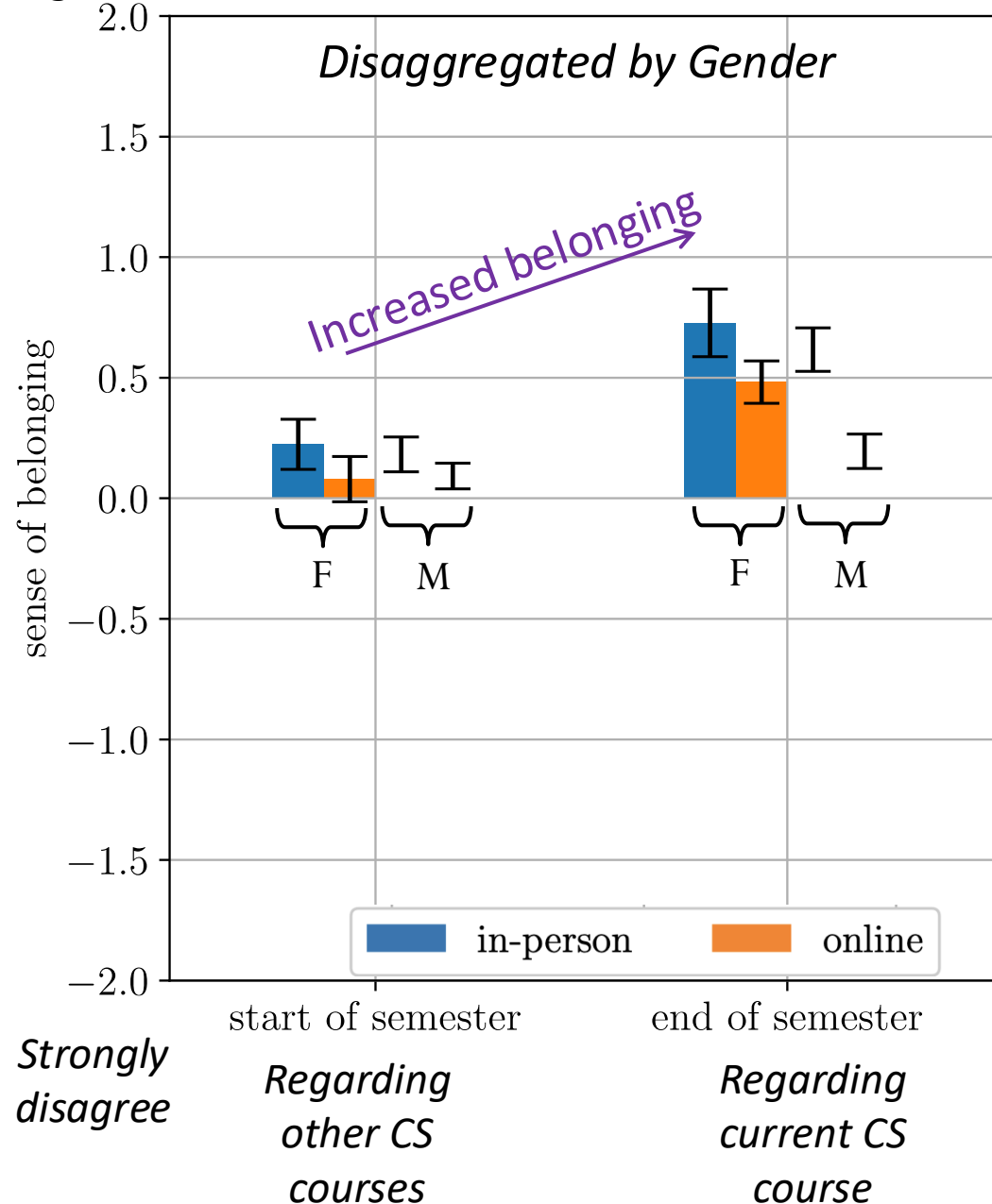
Survey questions regarding perceived comfort, support, and isolation

Increased SoB over the semester

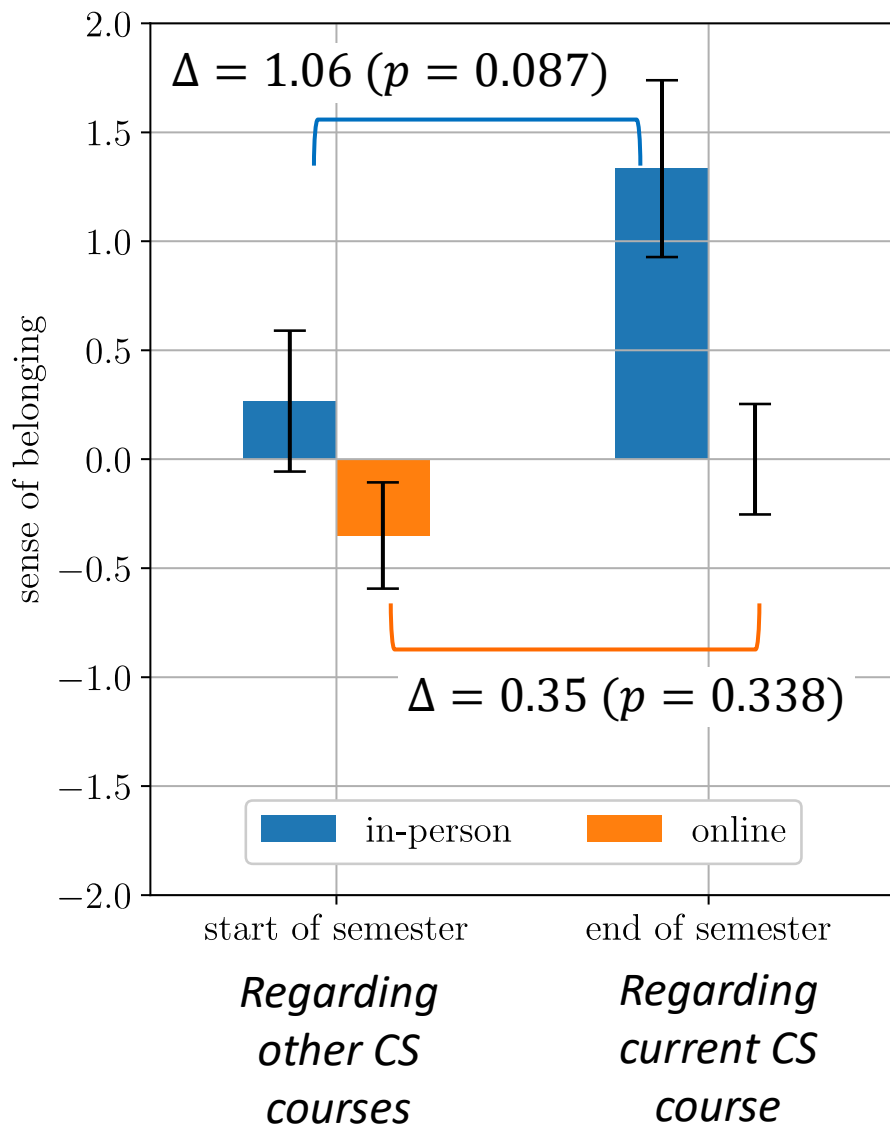
In-person students report **higher increase in SoB** when compared to **online** students

Online section: not significant increase in SoB for men; significant increase in SoB for women

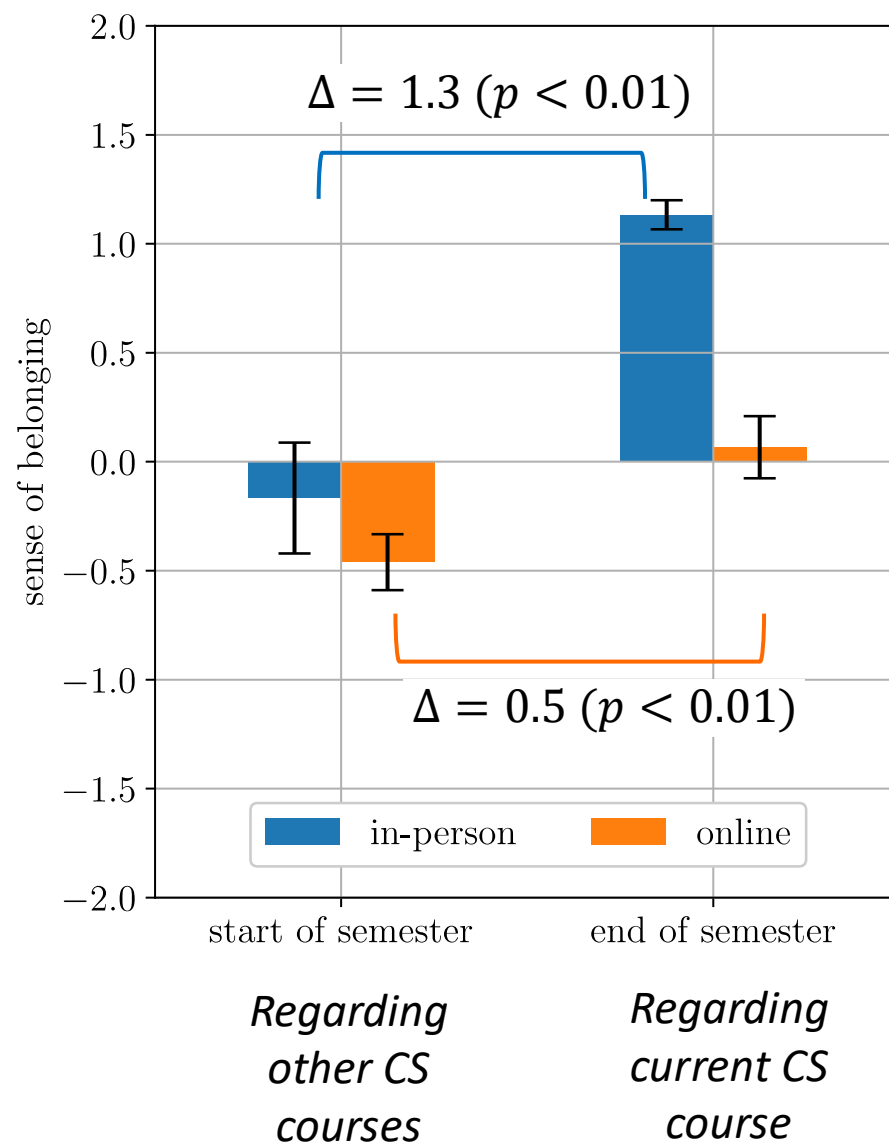
Strongly agree

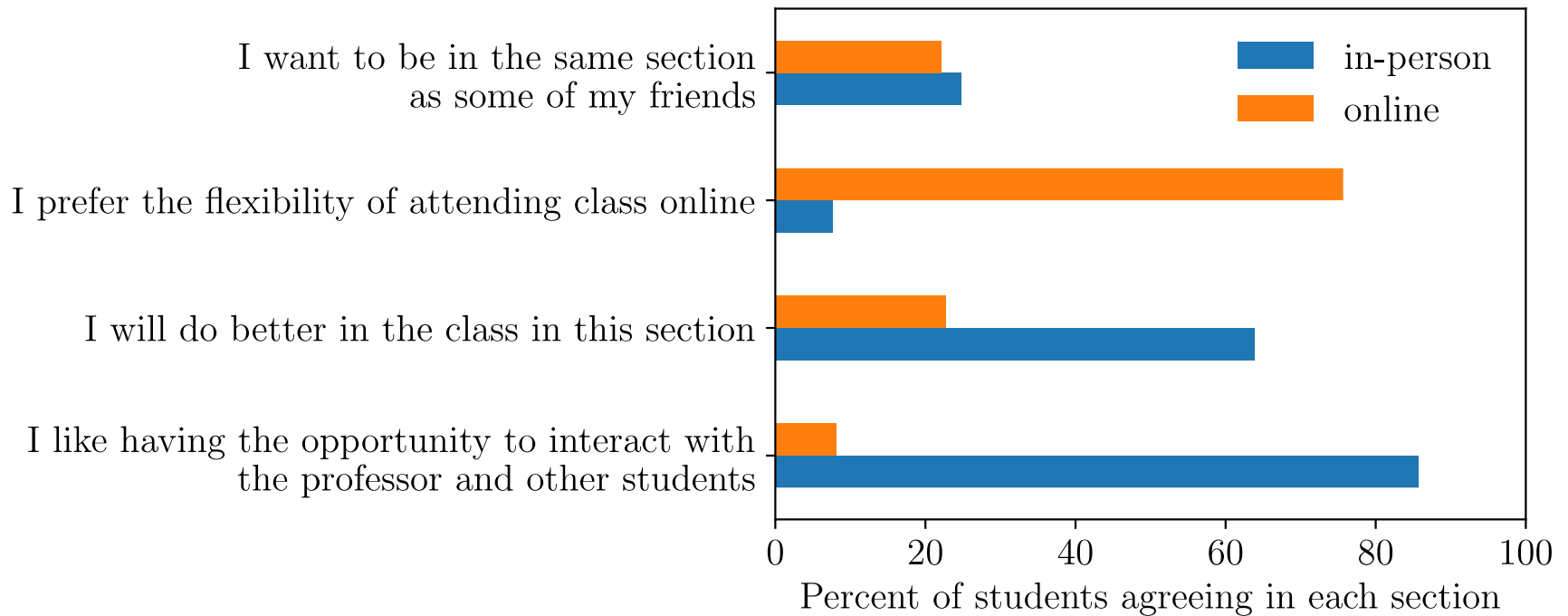


URM only



First-Generation only





Students select the section that best fit their preferences

Introduction and “Big Idea”

What are...

Numerical Methods ?

Numbers in a computer
(and how computer understands these numbers)



- Mathematical model
 - “algorithms” derived from math ideas to solve equations numerically
- Complexity of the problem
 - Slow vs fast
- Accuracy
 - Accurate vs inaccurate

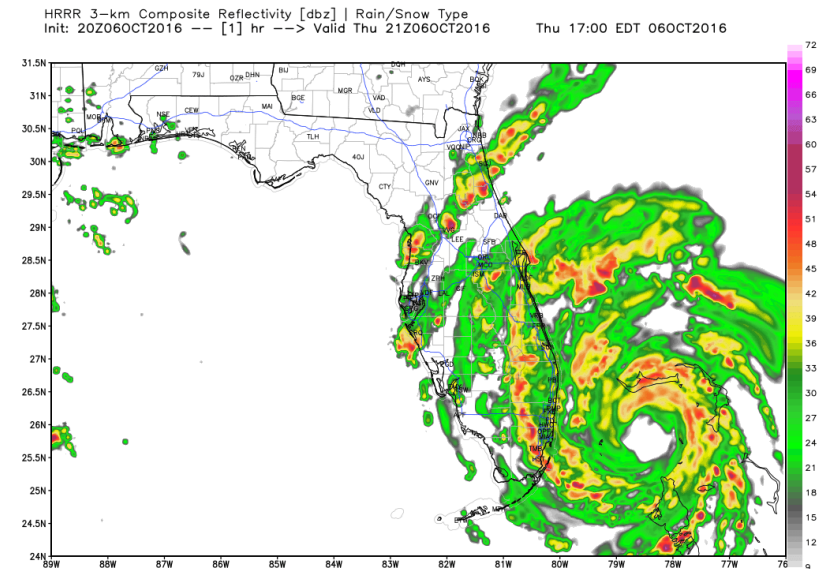
Method = Math + Complexity + Accuracy

Why is this course important?

1. Understanding and reconstruction of known problems
 - Natural disasters
 - Catastrophic failures
2. Prediction of unknown situations
 - Weather conditions
 - Behavior of new materials
3. Optimization of existing problems
 - Image recognition
 - Reduce fabrication costs



**Explosion of
Ariane 5 in 1996**



Goals for this course

- Understand how numbers are represented in the computer.
- When developing code, you will likely run into numerical errors. What are the sources of these errors?
- How can you avoid numerical errors?
- How can you choose a suitable algorithm for a given application?
- Use existing libraries to solve real applications.

(Numerical) **Method** = **Math** + Complexity + Accuracy

Mathematical model:

What equations can we use to represent our problem?

Accuracy:

Are we getting accurate results?

Why is the method not giving me the correct solution?

Complexity:

How long does it take to solve this problem?

Is it cost-effective?

Your entire CS 357
semester in a few
slides!

Are you ready?

Accuracy

- Why a numerical method might not give the right answer?
 - Computers have finite representation of numbers
 - Sometimes the “right answer” cannot be represented in a finite way
 - Example:

$$\pi = 3.1415926535897932384626433832795028841971\dots$$

Demo: Waiting for the number 1

```
from time import sleep

x = 0.0

while x != 1.0:
    x += 0.1
    print(repr(x))

    sleep(0.1)
```

What is going to happen when we run this code?

- A. Code will stop after printing 11 values for x
- B. Code will stop after printing 10 values for x
- C. Code will not stop
- D. Code will not start

Monte Carlo Methods

Texas Holdem Game: we would like to determine the probability of winning of a given starting hand

Physical experiment
vs
Numerical experiment



Numerical Experiments

- What do we want to know about a numerical experiment?
 1. What questions are we attempting to answer?
 2. What is the outcome of the experiment?
 3. Is it repeatable?
 4. Is the answer accurate?
 5. How long will it take?

Time vs accuracy trade-off

Question: Is running this method (with a certain accuracy) a good use of our time and/or computer resources?

Complexity

How long does it take to solve a problem?

Given A, B matrices of size $m \times m$, the matrix-matrix multiplication $A \cdot B$ takes τ seconds.

How long does it take to perform $C \cdot D$, matrices of size $2m \times 2m$?

```
from time import process_time
import numpy as np
from time import process_time
```

```
n = 2000
A = np.random.randn(n,n)
B = np.random.randn(n,n)

t = process_time() # store the time
C = A @ B
t = process_time() - t
print(t)
```

```
A = np.random.randn(2*n,2*n)
B = np.random.randn(2*n,2*n)

t2 = process_time() # store the time
C = A @ B
t2 = process_time() - t2
print(t2)
```

Linear system of equations: Image processing

How can we use linear operators to create blurred images? How can we do the inverse process?

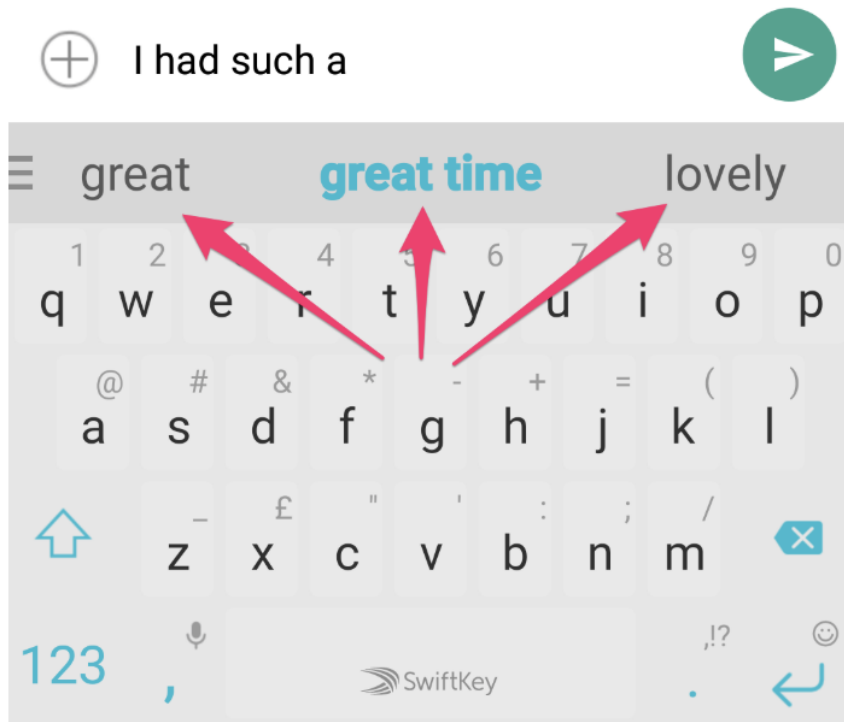


Image credit: <https://datacarpentry.org/image-processing/>

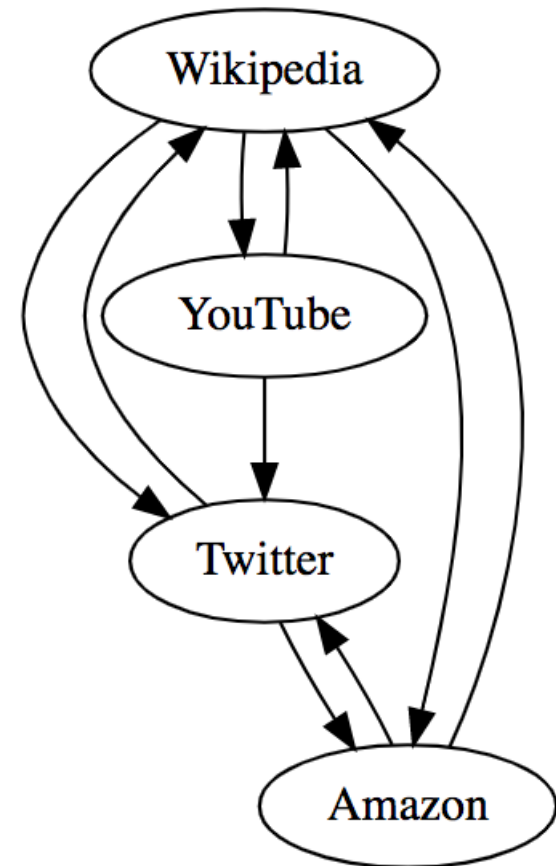


Markov chain

Word prediction

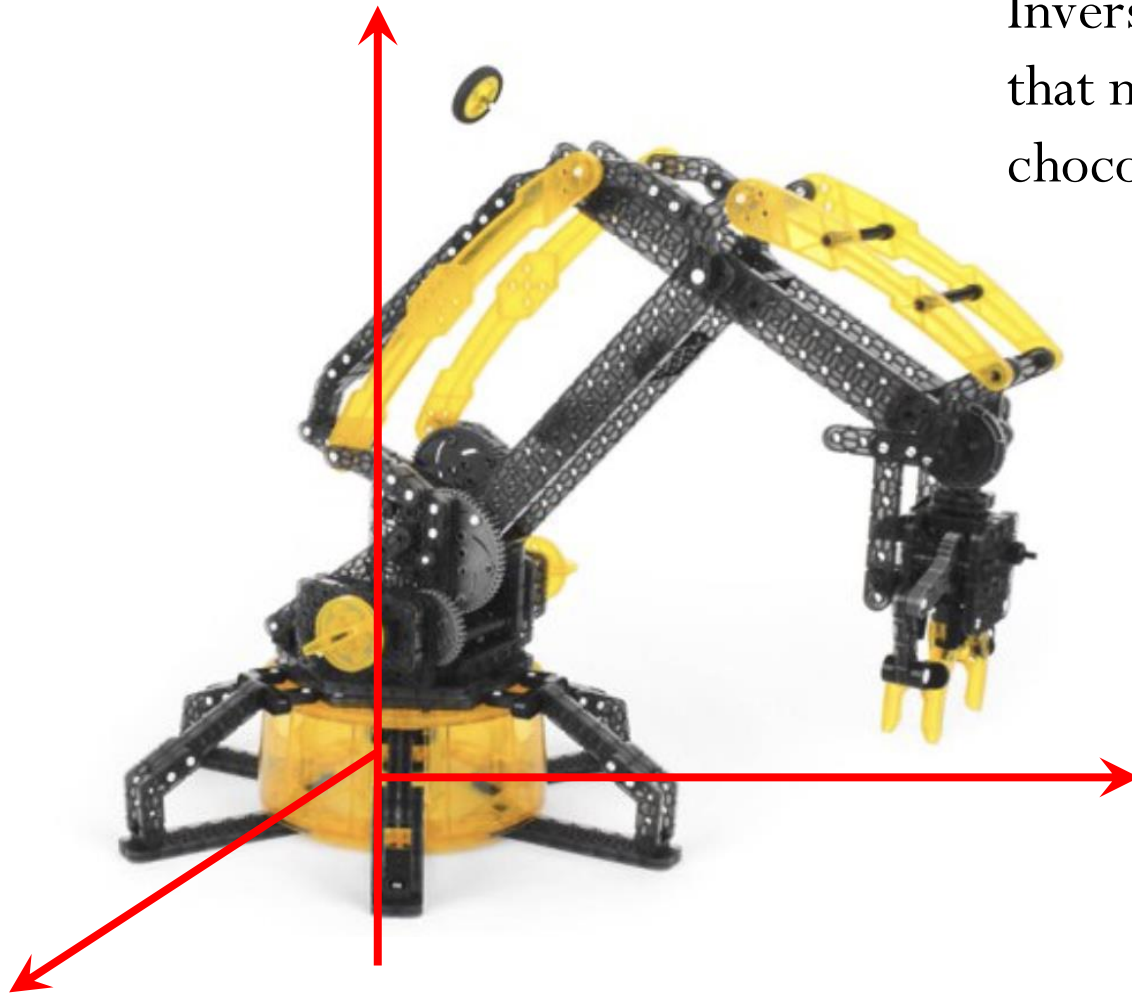


Page Rank



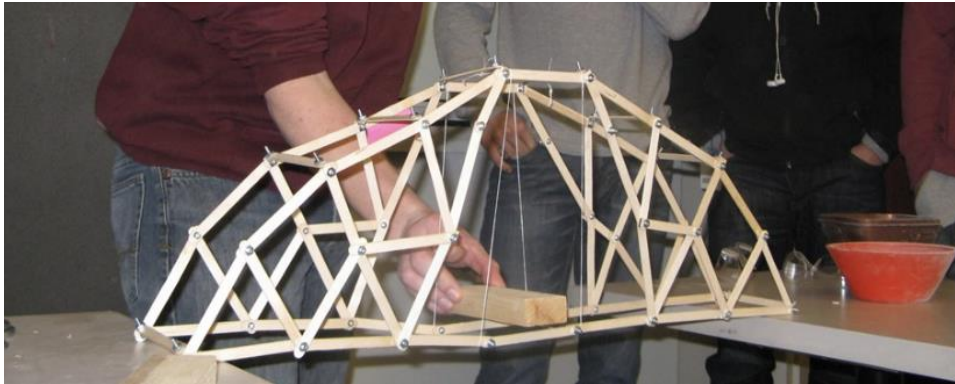
Nonlinear system of equations

Inverse kinematics: find the angles that make the robotic hand grab a chocolate candy!

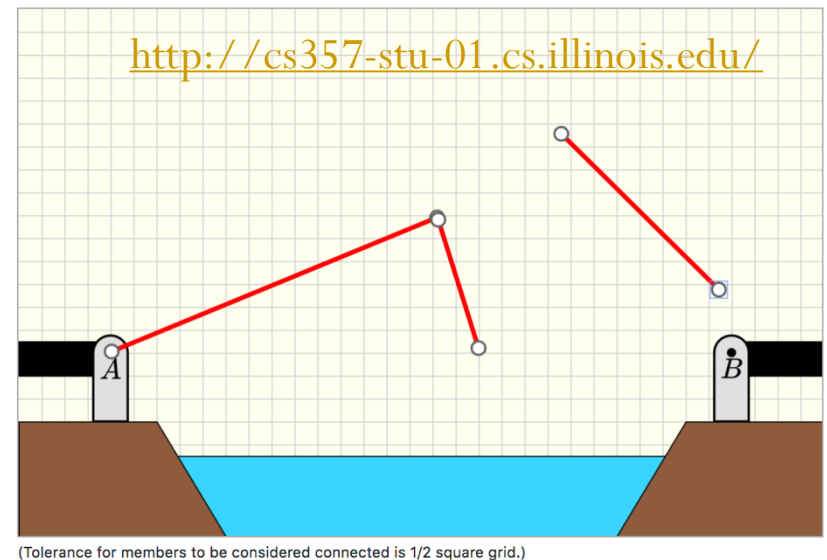
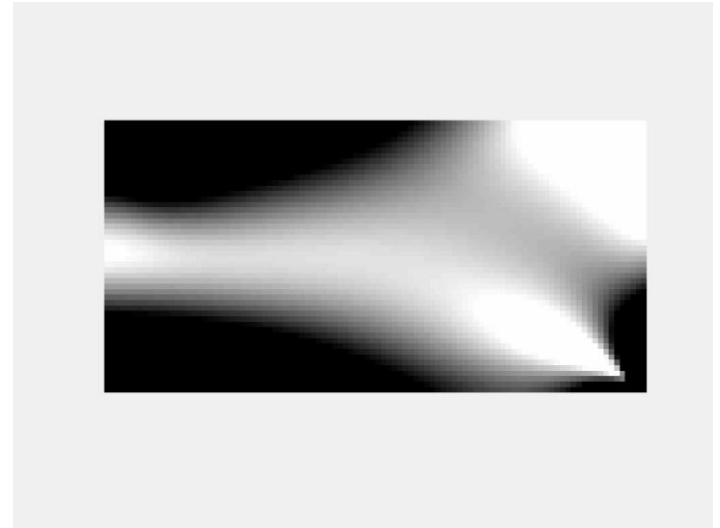


Optimization

Bridge design (high school projects)

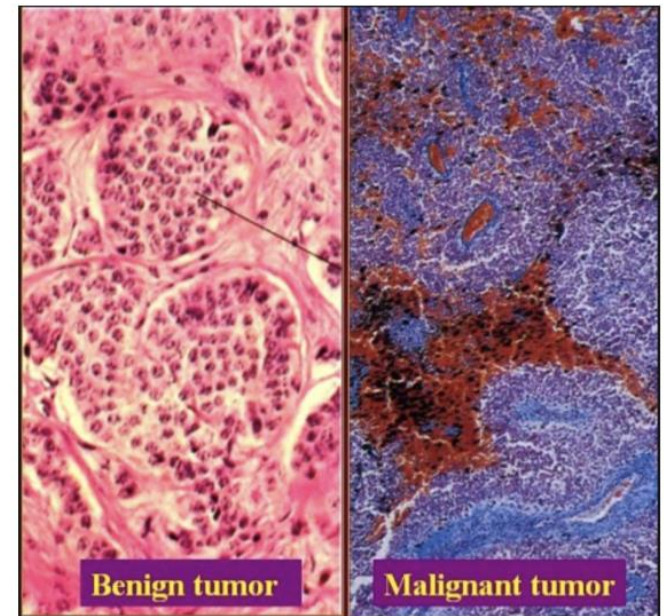
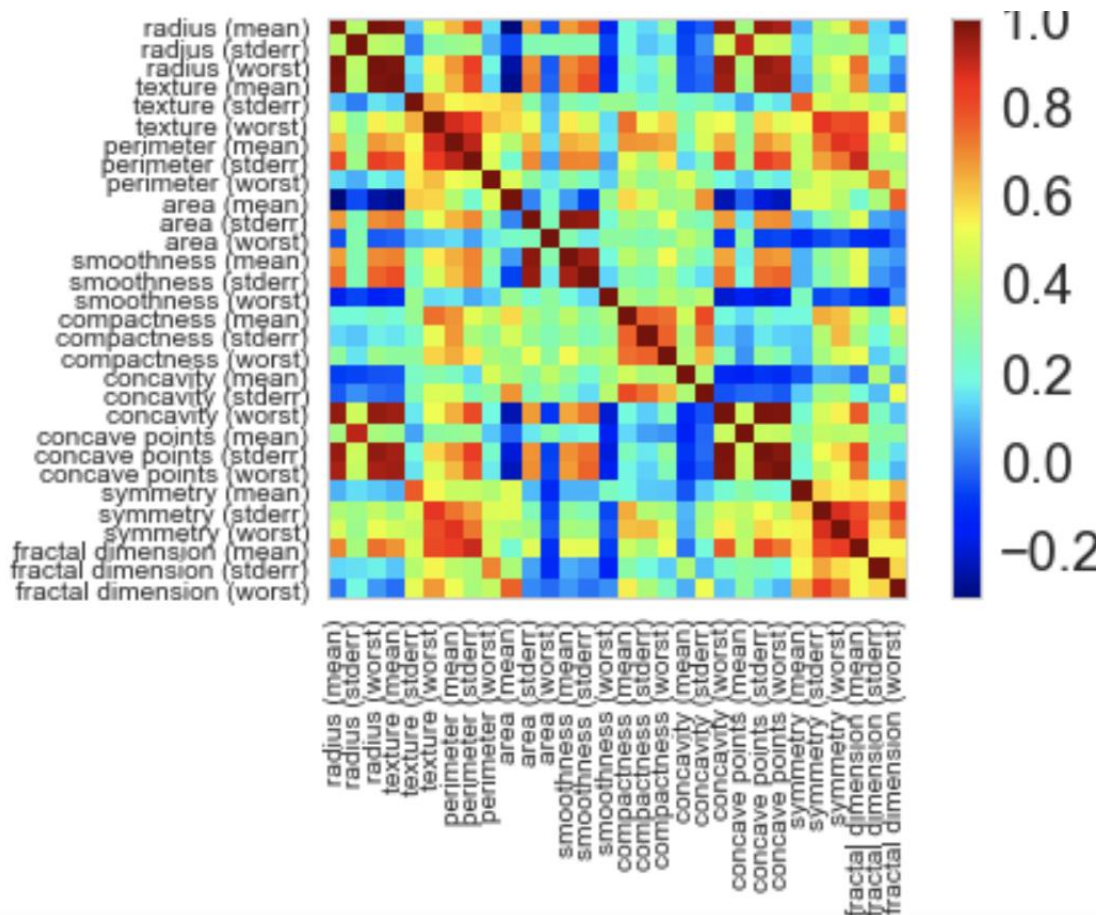


Numerical simulations to find optimized bridge designs



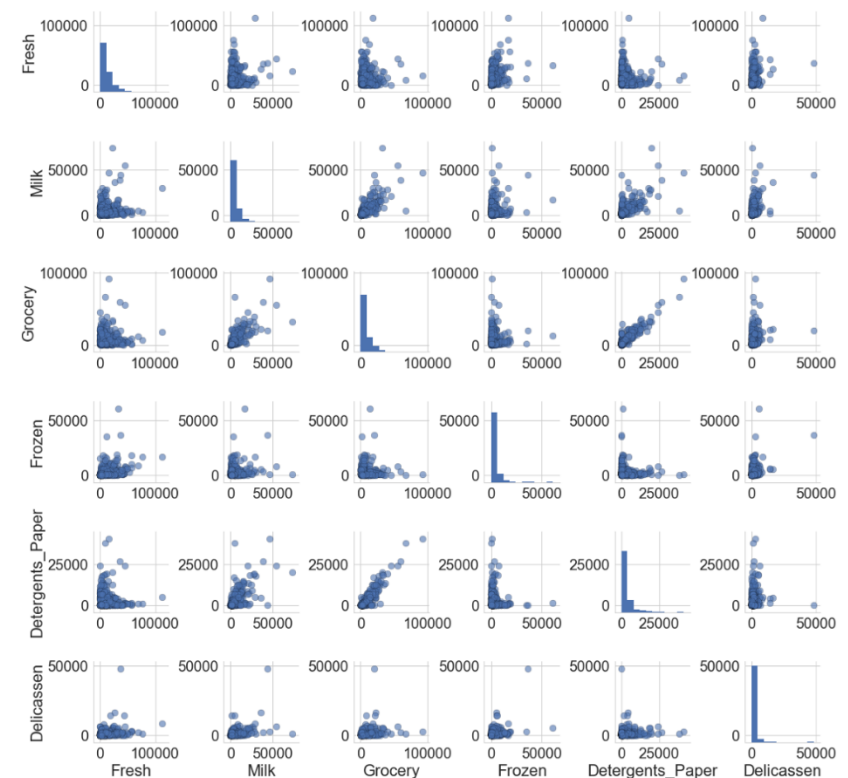
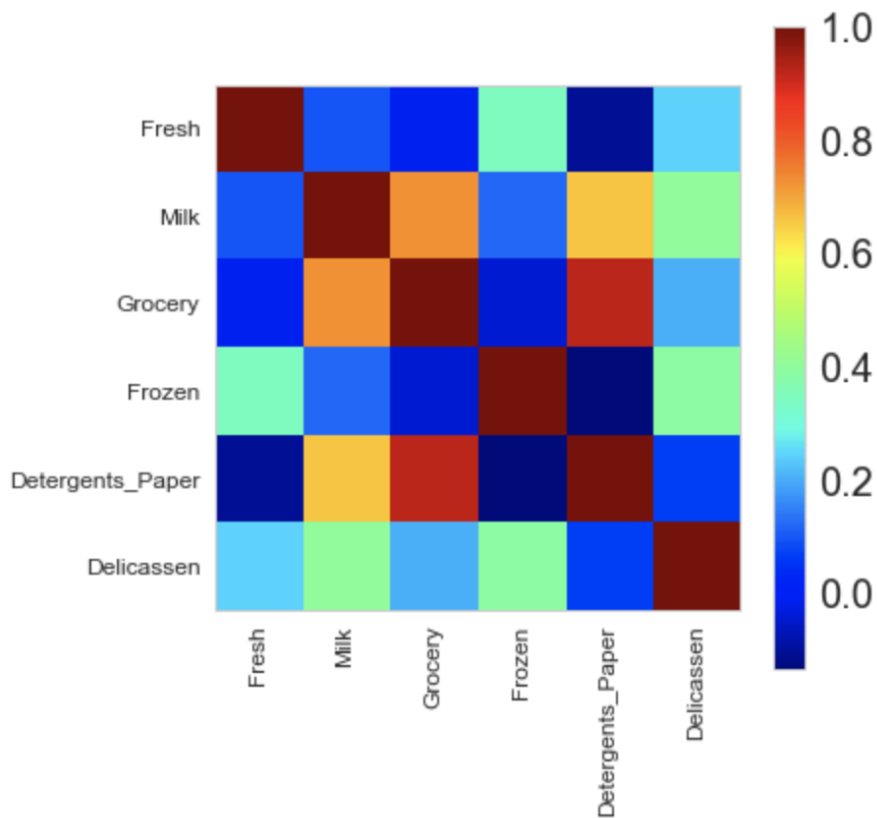
Linear Least Squares

Dataset containing the characteristics of cells for several patients. Can we make predictions if cells are benign or malignant?



Principal component analysis

Sometimes our dataset has too many features? How can we reduce the feature space and still keep the most important information?



WEEKS 1-2:

Both Tuesday and Thursday:

- Synchronous hybrid class for all students
- Section N at CIF
- Section M join on Zoom

STARTING FROM WEEK 3:

Tuesdays:

- Group activity
- Attendance required only for students in section N
- Section M students support via Discord
- Zoom for **mini-lecture** only

Thursday:

- **Optional** Study Hours
- Additional examples
- All students who want to get this additional help should go to CIF
- **No Zoom option**

Second day of classes...

Survey and consent form

Course surveys

S3

[Start of semester survey \(extra credit\)](#)

Quizzes

- 6 quizzes; lowest score is dropped (35% of final grade)
- 8-12 non-coding questions + 2-3 coding questions
- 50-min quizzes all at CBTF (check schedule on course website)
- DRES students should provide letter to CBTF
- We will use PrairieTest to make quiz reservations.
- MAKE YOUR RESERVATION AS SOON AS SLOTS OPEN! We will not provide special overrides because you were not able to find a slot that works for you.
- Practice Quizzes – generated mostly from the same set of questions that will generate the actual quizzes

Exams available for reservations			
Action	Exam	First date	Last date
Make a reservation	CS 357 (Sp23): Quiz 1	2023-01-30 00:01:00 (CST)	2023-02-01 23:59:00 (CST)

Assessments		
	Available credit	Score
Course surveys		
S3 Start of semester survey (extra credit)	None ⓘ	0%
Module 3. Errors and Big-O		
L3 Errors, Big-O notation, plots	100% until 11:00, Tue, Jan 27 ⓘ	Not started
HW3 Errors and Big-O	100% until 23:59, Thu, Jan 29 ⓘ	Not started
Module 2. Python		
L2 Introduction to Python	100% until 11:00, Tue, Jan 27 ⓘ	0%
HW2 Introduction to Python	100% until 23:59, Tue, Jan 27 ⓘ	0%
Module 1. Introduction		
L1 Introduction to CS 357 (NOT FOR CREDIT)	None ⓘ	Not started
HW1 Linear Algebra Review (WILL NOT COUNT FOR COURSE CREDIT)	100% until 23:59, Tue, Jan 27 ⓘ	Not started

Week 1

Tue, Jan 20, lecture 1

Syllabus and Intro to CS 357

Complete your asynchronous lecture today!

L1: Introduction to CS 357 (NOT FOR CREDIT)

L2: Introduction to Python

Also opening today:

HW1: Linear Algebra Review (NOT FOR CREDIT)

HW2: Introduction to Python

Due today:

Thu, Jan 22, lecture 2

Python Intro + mock group work

Complete your asynchronous lecture today!

L3: Errors, Big-O notation, plots

Also opening today:

HW3: Errors and Big-O

Due today:

L2: Introduction to Python

L2: Introduction to Python

Total points: 0/2

0%

Assessment is **open** and you can answer questions.

Available credit: 100% until 11:00, Tue, Jan 27 ⓘ

Resources: [Online Textbook](#), [Lecture Slides](#)

Question	Value	Variant history	Awarded points
Self-guided notebook (no pre-recorded video)			
L2.1. Prerequisite survey	1		— /1
L2.2. Python intro - self-guided notebook	1		— /1
Additional Python tutorials (optional)			
L2.3. Basic Python			
L2.4. Numpy			
L2.5. Matplotlib			

L2.2. Python intro - self-guided notebook

Open the workspace below and complete the IPython notebook.

 Open workspace

Select one of the answers below (there is no correct answer):

- ☐ (a) I completed the notebook, and I found it helpful.
- ☐ (b) I completed the notebook, but I did not find it helpful.
- ☐ (c) I did not complete the notebook because I already know how to use Python.
- ☐ (d) I did not complete the notebook (for other reasons).

Save & Grade *Single attempt*

Save only

Additional attempts available with new variants ⓘ

Collaborative Learning

Collaborative learning has been shown to improve:

- sense of belonging
- learning outcomes
- retention of knowledge

The benefits of collaboration can diminish:

- effort distribution is inequitable
- team members are stereotyped into specific roles (e.g., women expected to take notes)

The assignment of pre-defined roles can mitigate these issues

Collaborative Learning

Collaborative learning has been shown to improve:

- sense of belonging
- learning outcomes
- retention of knowledge

You will be encouraged to work with your peers in all “formative” assessments (Pre-lectures, HWs, MPs, GAs).

For Pre-lectures, HWs, MPs you must submit your individual work

Collaborative Learning

- Complete weekly activity in groups
- Week 1 and 2: randomly assigned groups via Zoom
- Then you will have the option to form your group
- You will also have the option to opt out and work alone, forfeiting participation points
- Starting from week 3: fixed groups

Supporting Collaborative Learning with Structured roles

Manager: keep team on task; enter the roles in PL

Recorder: enter most of the answers in PrairieLearn

Reflector: makes sure everyone is keeping up; complete survey



Consent was given for the media usage

When structured roles were required to alternate among group members...

- more equality in the work distribution among members
- groups scored better (on average a full letter grade
- groups completed work faster (on average 2.8 hours faster)

Meeting time preference (during class or another time on Tuesday)

- no significant effects on students' exam performance, sense of belonging or satisfaction regarding the course

Team consistency

- positive effects on students' exam performance and sense of belonging, but not on satisfaction

Using the Queue during GA

- To get support while on Zoom (Tuesdays 12:30pm-1:45pm)

 Enable notifications 

Close Queue

Show staff message

This affects anyone using this queue



On-Duty Staff

Mariana Teixeira Silva



Leave

Queue staff message



Please include the problem number - up to 2 questions each time you enqueue :)

Group Activites (for online section only)

Please put your **Zoom Breakout Room number** as your location.

Online Office Hours (Discord)

Please put your **Discord Voice Channel name** as your location. *Wait in your Discord Voice Channel until a staff member comes to help.*

In-Person Office Hours

These OH don't use the queue. Please find us in the Siebel Basement, and check Discord for any updates.

If you are working on a coding question, always try printing out the value you are unsure about and reading the stack trace carefully.

New question



Leo (tluo9)

breakout room 17 • 4 minutes ago

GA1 question 1.2

Start Answering!

Delete

Kajal (kpate457)

Zoom breakout room #14 • a minute ago

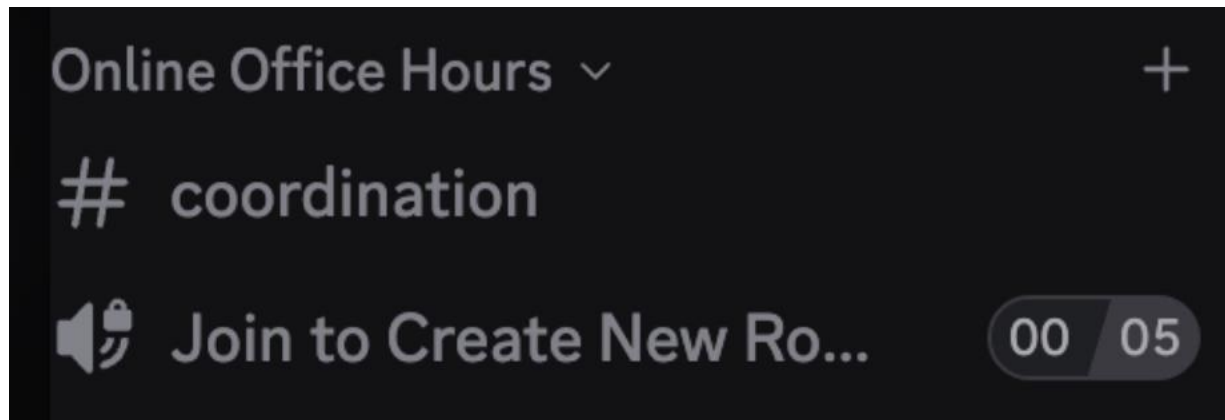
Question 1.2

Start Answering!

Delete

Using the Queue for Discord OH

- Click “Join to Create New Room” voice channel in the “Online Office Hours” category. You will be moved into your own room.



- Go to Queue and post a new question. For the location, type your Discord Voice Channel name.
- Use the #coordination channel if you have logistical question related to joining a voice channel

Creating a group assessment in PL

GA00

Workspaces for collaborative learning (NOT FOR CREDIT) 

GA00: Workspaces for collaborative learning (NOT FOR CREDIT) 

This is Group Activity 00: Workspaces for collaborative learning (NOT FOR CREDIT) for CS 357

This assessment can be done individually or in groups. A group must have no more than 3 students.
To work individually, you must also create a group, but you don't need to share your join code.

Group name

e.g. teamOne

Group names can only contain letters and numbers, with maximum length of 30 characters.

Create new group

Join code

abcd-1234

Join group

We will go over this first mock GA.



This is Group Activity 00: Workspaces for collaborative learning (NOT FOR CREDIT) for CS 357

Group name: mygroup**Join code:** mygroup-3786

This assessment can be done individually or in groups. Use your join code if you wish to invite others to join the group. A group must have no more than 3 students.

[Leave the Group](#)**Group members:**

- mfsilva@illinois.edu - Manager

Your group's role configuration is currently invalid. Please review the role requirements and assign a valid role configuration. Question submissions are disabled until the role configuration is valid.

▼ Manage group roles **2**

1 more student needs to be assigned to the role "Recorder" (0 assigned, 1 expected).

1 more student needs to be assigned to the role "Reflector" (0 assigned, 1 expected).

This assessment contains group roles, which selectively allow students to view questions, submit answers, and change group role assignments.

User	Roles
mfsilva@illinois.edu	☒ Manager ☐ Recorder ☐ Reflector ☐ Contributor

[Update Roles](#)

You will now complete this GA with a random group. We will give you 7 minutes.

Creating the groups

Course surveys

S1

Select your group (NOT FOR CREDIT)

Submit the survey by next Friday!

Students with time conflict override (not able to attend lecture) **MUST** find a team that agrees to meet at a different time.

Fixed groups start on week 3

If you know 2-3 other students taking CS 357 this semester, and you have agreed to complete the group activities together, you can request to be placed in the same group.

To submit this request, your group must select a group name, so that all members can submit the same answer below:

In the entry field below, enter your group's selection for the **group name**.

group name:



Important notes:

- Every student that enters the same group name will be placed in the same group.
- Make sure you agree on a creative and **unique group name**. For example, you can use the members last names combined. You don't want to be placed in the wrong group by mistake.
- **Groups must have 2-3 students.** If more than 3 students or less than 2 students submit this request using the same group name, ALL these students will be placed in groups at random!
- **Groups can only be formed with students registered in the same section.**

Students who do not submit this survey will be placed at a group at random. Students who are assigned to a random group in the online section must attend the Zoom meeting during class time on week 3 (they can make other arrangements for future weeks).

If you choose to opt out of a group, you will forfeit your participation grade. Read more details about participation grades [here](#).

- ☐ (a) Put me in a group please! I'm ready to commit to working with my group.
- ☐ (b) ONLINE (section M) only: I am opting out of a GA group!

Select all possible options that apply.

If you are in the in-person section, write down your desired table number:

table number:

integer



Completing the GAs (Section N)

- Attendance at CIF 35 is required
- We will use the QR code from your Illinois App
- You must not be late
- Absences:
 - If you are ill, do not come to the classroom:
 - Make arrangements with your team to connect remotely
 - Submit a request for excused absence

DO NOT JOIN A GA IF YOU ARE NOT PARTICIPATING IN THE ACTIVITY (not in the classroom, or remote attendance due to illness). You will receive a zero in that GA.

Completing the GAs (Section M)

IF YOU PRE-SELECTED YOUR TEAM

- Your team can agree on a time and location to meet

IF YOU WERE ASSIGNED TO A RANDOM TEAM

- Your team must meet via Zoom during class time on week 3. You can make other arrangements for other weeks

TO GET SUPPORT

- GA support offered 12:30pm to 1:45pm via Zoom
 - Queue must be used to request help
- Only team members participating in the completion of the GA can use the Join code.

How the final GA score is computed

GA score (from PrairieLearn): counts towards 9% of the grade (drop of 2 lowest scores, start counting from week 2 (GA1))

Participation score: counts towards 2% of the grade (drop of lowest score, start counting from week 3 (GAF2))

	In-person	Online
Participation = 1	Attended class	Completed feedback survey
Participation = 0	Did not attend class	Did not complete feedback survey

Students who complete GA individually will not receive participation points.

Difference between Sections M and N

Section N (in-person)	Section M (online)
Attendance is required	Attendance not required
Start GA during class time. Most groups will complete GA before the end of the class. Can still complete GA until 10pm	Can start GA at any time from 12:30pm and complete by 10pm.
Support during class time at CIF 35.	Support during class time via Zoom.
A student can request remote attendance in case of sickness. The remote arrangements should be decided by the group. Remote attendance will be verified by course staff during class time.	Students can make arrangements for remote attendance when someone is sick without communication with course staff (in case a group meets in-person)

Practice Group Activity

GA 0

Get started with GAs (NOT FOR CREDIT) 

Extra credit questions:

The extra credit question(s) will be later graded by a Large Language Model (LLM) using a rubric provided by the instructor. Once the grading is completed, please review the rubric items selected by the LLM. While the LLM is designed to follow the rubric, AI tools can occasionally make mistakes. If you believe your response was graded incorrectly, you may submit a review request using [this form](#). Your submission will be reviewed by a course staff, and your score will be manually adjusted if necessary.

This is just a mock GA to help you get familiar with how to work in groups and use Jupyter Notebooks for completing GAs. This GA does not count towards your course grade (and so no extra credit points will be awarded).

Question	Auto-grading			Manual grading points	Total points
	Value	Variant history	Awarded points		
Getting to know your group mates					
GA 0.1. Icebreaker	4		— /4	—	— /4
Here is how we use Jupyter Notebooks to complete GAs					
GA 0.2. Getting familiar with workspace 	50		— /50	—	— /50
To help reviewing linear algebra and intro to programming					
GA 0.3. Creating a matrix 	12		— /12	—	— /12
GA 0.4. Use NumPy to solve a system of equations 	12		— /12	—	— /12
GA 0.5. Solution to heat equation 	12		— /12	—	— /12
Helpful notebook to estimate final scores in CS 357					
GA 0.6. Computing the final score 	10		— /10	—	— /10
Extra credit questions (AI-graded)					
GA 0.7. Identify one bug in this code 	—		—	— /2	— /2
GA 0.8. Identify the bug in this code 	—		—	— /1	— /1